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3 October 1984
SUPERSEDING
MIL-I-25437A(USAF)
14 November 1967

MILITARY SPECIFICATION

INDICATORS, AIRCRAFT, PRESSURE, VOLTAGE RATIO TYPE, GENERAL SPECIFICATION FOR

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 This specification covers the general requirements for voltage ratio type pressure indicators that operate on alternating current.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specification and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-636	Box, Fiberboard

MILITARY

MIL-P-116	Preservation-packaging, Methods of
MIL-C-5015	Connectors, Electrical, Circular Threaded, AN-Type, General Specification For
MIL-C-5541	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-A-8625	Anodic Coatings, For Aluminum and Aluminum Alloys
MIL-T-25435	Transmitters, Pressure, Aircraft, Variable, Reluctance Type, General Specification For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: ASD/ENES, Wright-Patterson AFB, OH 45433 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6685

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STANDARDS

DOD-STD-100	Engineering Drawing Practice
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection Of
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-889	Dissimilar Metals
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, With Appropriate Test Methods
MS 33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form Of
MS 33638	Case, Instrument, Flange-Mounted, Aircraft
MS 33639	Case, Instrument, Clamp-Mounted, Aircraft
MS 33678	Connector, Receptable, Electric, Integral Mounting

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein.

UNIFORM CLASSIFICATION COMMITTEE, AGENT RULER

Uniform Freight Classifiction

(Application for copies should be addressed to Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Type of pre-production testing. The type of pre-production testing (qualification or first article) to be accomplished on indicators shall be as specified in the detail specification.

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials

3.3.1 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the indicator except where magnetic materials are essential.

3.3.2 Nonferrous materials. Nonferrous materials shall be used for all parts of the indicator except where ferrous materials are essential.

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3.3.3 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt fog, or atmospheric conditions likely to be met in storage or normal service.

3.3.4 Dissimilar metals. Unless suitably protected against electrolytic corrosions, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.3.5 Toxic and corrosive fumes. The indicator shall be of such material and construction that fumes, when burned, or gases that combine with the atmosphere to form acid or corrosive alkali will not be liberated.

3.3.6 Protective treatment. When materials are used in the construction of the indicator that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.4 Design and construction. The indicator shall be designed to provide remote indication of aircraft functions when corrected to a transmitter conforming to MIL-T-25435. External wiring of the indicator shall be in accordance with figure 1. The indicator shall be of the voltage-ratio type which operates a pointer that continuously indicates on a dial the measured pressure.

3.4.1 Cap. A separate nonhermetically sealed cap may be used on the back of the case for installing the electrical receptacle. The cap shall be made of nonferrous, low-density metal.

3.4.2 Case to cover glass distance. The distance between the outer surface of the cover glass and the front edge of the case shall be the minimum practicable and shall not exceed 1/32 inch.

3.4.3 Dial. The dial shall be in accordance with the detail specification. Dials installed in cases conforming to MS36639 shall have a minimum dial diameter across the outside ends of the graduations of 1-3/4 inches and dials installed in cases conforming to MS33638 shall have a minimum dial diameter across the outside ends of the graduations of 1-15/16 or 1-3/4 inches, as applicable.

3.4.3.1 Dial to cover glass distance. The distance between the indicator dial and the inside of the cover glass shall be the minimum practicable and shall not exceed 0.140 inch.

3.4.4 Filler tube. A tube or other facility for evacuating and filling the case shall be provided at the rear of the case and shall be recessed beneath the level of the back plate surface. If a tube is used, it shall be of such length as to permit three refillings of the case.

3.4.5 Cover glass. The cover glass shall be clear, flat, and free from flaws that would interfere with the normal reading of the instrument. The thickness of the cover glass shall be the minimum practicable and entirely suitable for the purpose.

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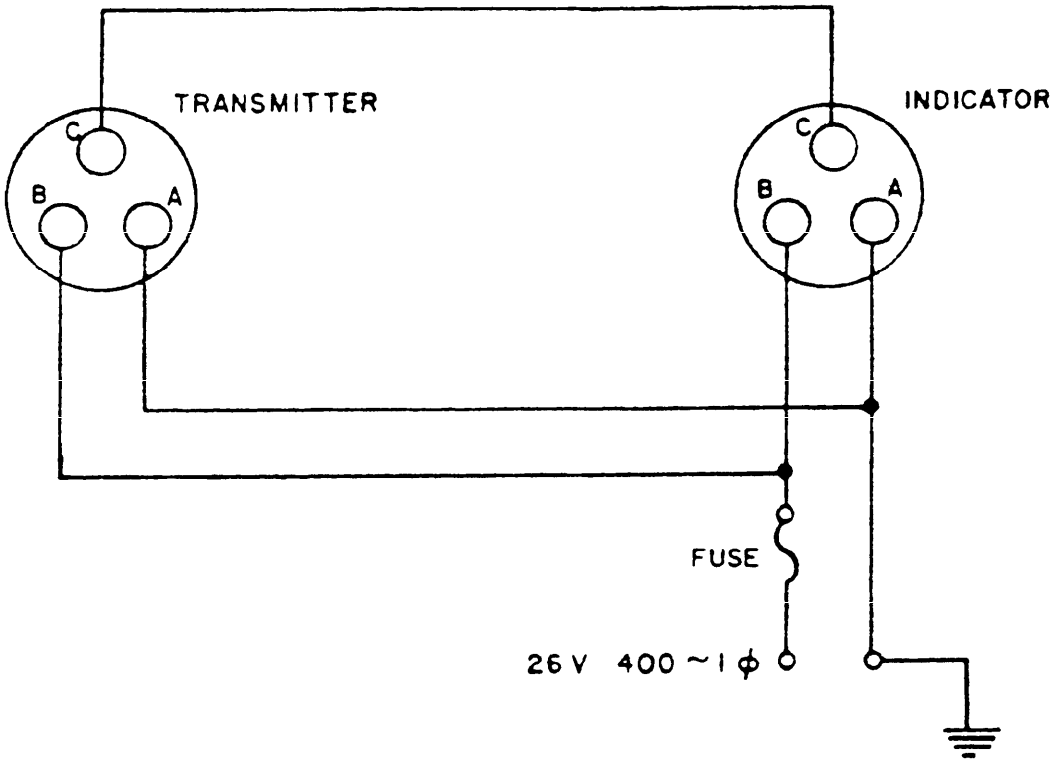


FIGURE 1. External wiring diagram.

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3.4.6 Fasteners. The dial shall be securely fastened to the frame of the mechanism in such a manner that it will not loosen or turn when the indicator is vibrated. If screws are used for attaching the dial, they shall be so located that they will not interfere with dial markings or prevent the markings from being located in the proper position.

3.4.7 Filling medium. The filling medium shall be of at least 98 percent purity, free of dust particles, and shall contain no more than 0.006 milligram of water vapor per liter (dewpoint -65°C) at the filling pressure. The filling medium shall be 88 to 92 percent nitrogen and the remainder helium. The absolute pressure of the filling medium in the case shall be 12 to 14.7 psi. Other concentrations of filling medium shall not be used except with prior approval of the procuring activity.

3.4.8 Power supply. The indicator shall operate from a 26 ±2V, 400 ±10 Hz single-phase power source. The wave form of the a-c voltage is essentially a sine wave with a crest factor of 1.41 ±10 percent and with harmonics having respective amplitudes of less than 5 percent of the fundamental. The indicator shall not be damaged if a voltage of 10 percent greater than the maximum value specified is applied.

3.4.9 Electrical connectors. Electrical connectors shall be in accordance with MIL-C-5015. The connector shells shall be protected against corrosion and shall meet the requirements specified herein. Each hermetically sealed indicator shall be provided with an MS33678-10SL-3P receptacle.

3.4.9.1 Polarizing key. The electrical connector polarizing key shall be located within ±20° of the vertical centerline of the side nearest the top of the indicator.

3.4.10 Internal wiring. Internal wiring shall be color coded and shall be connected as shown on figure 2.

3.4.11 Case. Unless otherwise specified in the detail specification, the case shall consist of a body and bezel ring. The case shall be sufficiently rugged so that small distortions will not affect the accuracy of the indications.

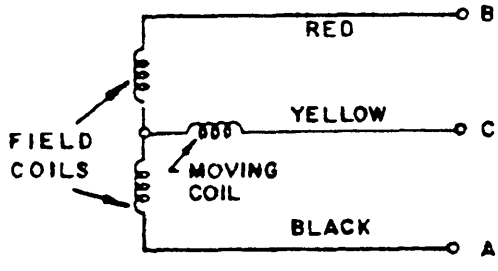
3.4.11.1 Bezel ring. The bezel ring shall be made of nonferrous, low density metal and shall have a durable dull-black finish. It shall be so constructed that it will properly position the cover glass.

3.4.11.2 Hermetically sealed cases. The case shall provide a hermetically sealed enclosure for the indicator internal mechanism. It shall be so designed that the internal mechanism may be removed from the case, replaced, and the case resealed without the use of special tools unless they are approved by the procuring activity. The hermetic sealing shall be so accomplished that the seal will not be dependent upon materials that will be affected by any action to which the indicator may be subjected.

3.4.12 Dial markings. Numerals and letters on the dial shall conform to MS33558 and shall be so durable as to withstand usage encountered in service.

3.4.12.1 Numerals shall distinctly indicate the graduation to which each applies. When several numerals are used in a group, the space between each numeral shall be not less than 1/64 inch.

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A-B - POWER 26V 400 ~1ϕ
A-C - TAP VOLTAGE

FIGURE 2. Internal wiring diagram.

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3.4.12.2 Pointers, dials, and dial markings shall be finished in the color specified in the detail specification.

3.4.13 Pointers. Pointers shall be in accordance with the detail specification.

3.4.13.1 Pointer return. In the event of power failure, the pointer shall position itself below the minimum scale point. The displacement of the pointer below the minimum scale point shall be equivalent in degrees to at least two percent of full scale, but not so great as to be misinterpreted as an overscale reading.

3.4.14 Pivots, bearings, and gears shall neither bind nor shake and shall be as nearly frictionless as practicable.

3.4.15 The indicator shall be so constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipment, storage, installation, and service.

3.5 Performance. The indicator shall operate as specified herein after subjection to the following conditions:

a. Temperatures: Temperatures ranging from -65°C to +71°C

b. Vibration: Vibration of 2g's from 32 to 500 Hz

c. Temperature-altitude: Pressure altitude of 100,000 feet at -55°C

d. Thermal shock: Eight cycles of alternate immersions in tap water at 85°C ±2°C and 5°C ±2°C.

e. Salt fog: Exposure to salt sea atmosphere for 48 hours without evidence of corrosion or deterioration.

3.5.1 Tolerances. The indicator shall operate within the tolerances specified in the detail specification.

3.5.2 Dielectric strength. The indicator shall withstand 200V rms at a commercial frequency for a period of 5 seconds.

3.5.3 Case leakage. When case leakage is checked with a mass spectrometer-type helium leak detector, leakage shall not exceed 0.1 micron cubic foot per hour. When the case is immersed in water, there shall be no leakage.

3.5.4 Fogging. There shall be no evidence of moisture or oil fog on the cover glass when the indicator is subjected to a temperature of 85°C ±5°C for a minimum period of 2 hours and the face of the cover glass immediately immersed in an ice bath for 3 minutes.

3.5.5 Damping. The time required for the indicator to properly respond to a step input equivalent to a 60-psi pressure change shall not exceed 2.0 seconds with not more than two overshoots.

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3.5.6 Electrical characteristics. The indicator shall have the following electrical characteristics with a normal excitation of 26V, 400-cycle, single phase applied across terminals A and B.

3.5.6.1 Primary circuit. With terminal C open circuited, the current drawn by the indicator shall not exceed 100 ma.

3.5.6.2 Power input. The indicator power input shall not exceed 0.60 watts.

3.5.6.3 Tap voltage. The maximum open-circuit voltage between terminals A and C and between B and C shall not exceed 18V.

3.5.7 Effect of lead length. When the lead length between the indicator and transmitter is changed from 5 to 300 feet, the indication shall not change more than 1 psi.

3.5.8 Magnetic effect. The compass deflection shall not exceed 1 psi when the indicator is positioned 5-1/2 inches from the compass and the horizontal magnetic field intensity at the indicator is 0.17 to 0.19 oersted.

3.6 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of DOD-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.7 Finishes and protective coatings

3.7.1 Aluminum alloy parts. Aluminum alloy parts shall be coated with an anodic film conforming to MIL-A-8625 except that small holes, straight threads, and case inserts need not be anodized. If such parts are contained in a hermetically sealed enclosure, they need not be anodized.

3.7.1.1 Aluminum alloy parts that do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

3.7.1.2 When abrasion resistance is a factor, chemical films in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

3.8 Screw threads. Unless otherwise specified, the threads of all machine screws shall conform to MIL-S-7742.

3.9 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.9.1 Nameplate. The indicator shall be marked for identification in accordance with MIL-STD-130.

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3.10 Workmanship. The indicator shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, welding, riveting, brazing, plating, painting, machine screw assemblies, and freedom of parts from burrs and sharp edges.

3.10.1 Dimensions and tolerances. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability or performance of the indicator, they shall be held or limited accordingly.

3.10.2 Cleaning. The indicator shall be thoroughly cleaned and loose, spattered, or excess solder, metal chips, or other foreign material removed during and after final assembly. Burrs, sharp edges, and resin flash that might crumble shall also be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as quality conformance inspections. The type of preproduction testing shall be as specified in the detail specification.

4.3 Inspection conditions

4.3.1 Standard atmospheric conditions. Wherever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

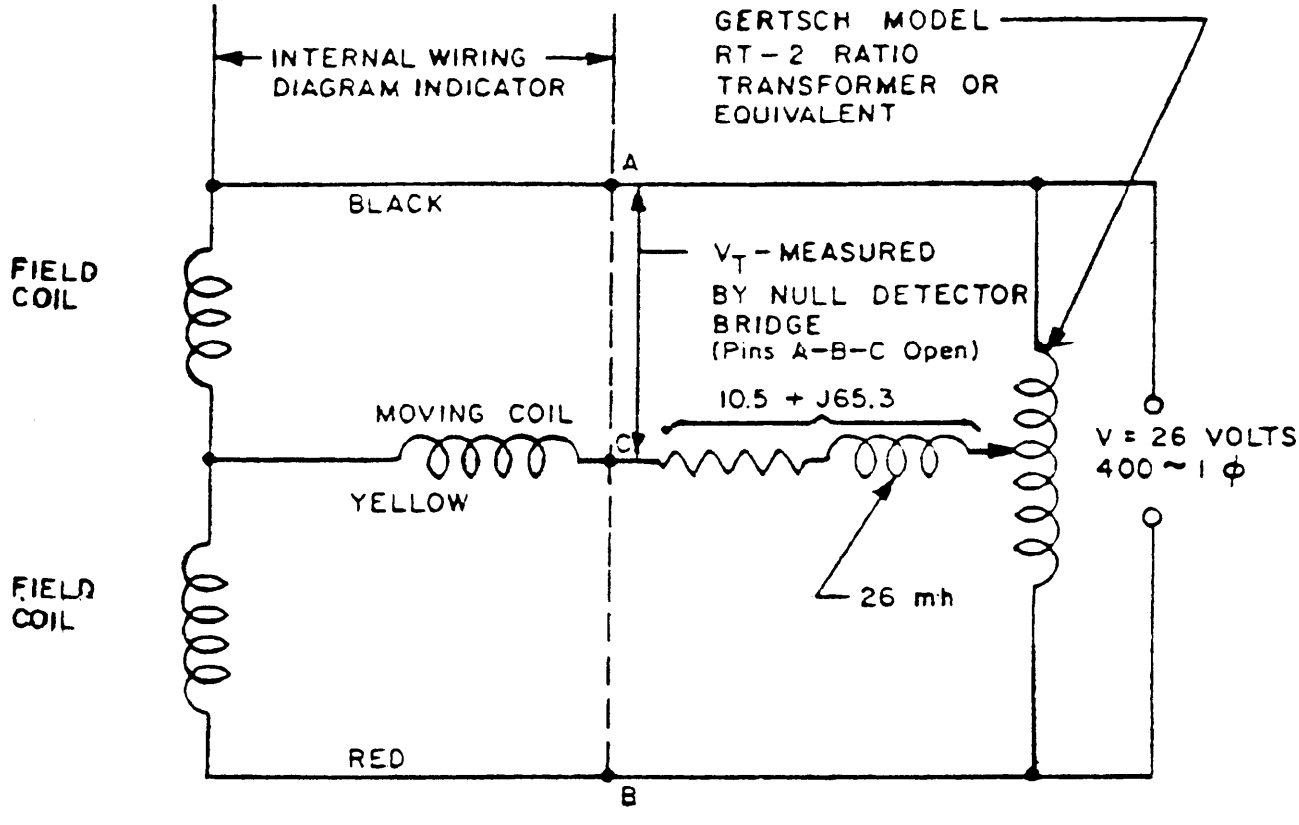
4.3.2 Attitude. Unless otherwise specified, the indicator shall be tested in the normal operating position with the plane of the dial vertical.

4.3.3 Tapping. Unless otherwise specified, the indicator shall be lightly tapped or vibrated before each test reading is taken.

4.3.4 Test voltage. The indicator shall be tested with an applied voltage of 26.0V ±0.5V single-phase 400 ±5 Hz a-c power.

4.3.5 Test standard. A standard ratio transformer circuit conforming to figure 3 shall be used for testing the indicator.

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Approximate Indication (Percent Full Scale) = $305-19.6V_T$ (Volts RMS)

FIGURE 3. Test stand circuit.

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4.4 Quality conformance inspections. Quality conformance inspections shall consist of:

a. Individual tests

b. Sampling plans and tests.

4.4.1 Individual tests. Each indicator shall be subjected to the following tests as described under 4.5:

a. Examination of product

b. Dielectric strength

c. Scale error and friction

d. Position error

e. Sealing.

4.4.2 Sampling plans and tests

4.4.2.1 Lot. A lot shall consist of indicators manufactured under substantially the same conditions and submitted for acceptance at essentially the same time.

4.4.2.2 Sampling plan A. Unless otherwise specified, 3 indicators shall be selected at random from the first 15 produced on the contract or order and subjected to the following tests as described under 4.5:

a. Electrical characteristics

b. Magnetic effect

c. Effect of lead length

d. Temperature-altitude

e. Low temperature exposure

f. High temperature exposure

g. Thermal shock

h. Salt fog

i. Damping.

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4.4.2.3 Sampling plan B. One indicator shall be selected at random from each lot of 100 or less on the contract or order and subjected to the following tests as described under 4.5:

a. Vibration

b. Low temperature

c. High temperature

d. Fogging.

4.4.2.4 Rejection and retest. When one or more items selected from a lot fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of the failure have been determined and appropriately corrected.

4.4.2.5 Individual tests may continue. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or later produced shall not be made until it is determined that items meet all the requirements of the specification.

4.5 Methods of inspection

4.5.1 Examination of product. The indicator shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and markings. Prior to assembly, dials and pointers shall be inspected as to markings, dimensions, and finishes.

4.5.2 Dielectric strength. A potential of 200V rms at a commercial frequency shall be applied between any pair of points for 1 minute. If the design of the indicator is such that there exists between the test points a number of electrical circuit elements that form a conducting path for alternating current and would be damaged by the test voltage, the elements shall be removed, open circuited, or short circuited prior to application of the voltage. There shall be no insulation breakdown as a result of this test.

4.5.3 Scale error and friction. The standard transformer circuit shall be adjusted to provide voltages across terminals B and C corresponding to the pressure specified in the detail specification. Two readings shall be taken; the first before, and the second after the instrument is tapped. The difference between the first and second readings shall not exceed the friction tolerance specified in the detail specification. The second reading shall not exceed the scale error specified in the detail specification.

4.5.4 Position error. The standard transformer shall be adjusted to provide a voltage across pins B and C corresponding to the pressure specified in the detail specification. The reading of the indicator taken while the indicator is in a normal upright position shall not differ from the reading while the indicator is held in any other position by more than the amount specified in the detail specification.

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4.5.5 Sealing. When a nitrogen-helium mixture is used as the filling medium, a mass spectrometer-type of helium leak detector shall be used to test for leakage. The leak rate shall not exceed 0.1 micron cubic foot per hour. When helium is not used as a part of the filling medium, the indicator shall be immersed in deaerated distilled water treated by adding approximately 0.1 percent by weight of alkyl-aryl-sodium-sulfonate type wetting agent (40 percent active ingredient). The absolute pressure of the air above the water shall be reduced to approximately 2-1/2 inches Hg. Any bubbles coming from within the instrument case shall be considered leakage and shall be cause for rejection. Bubbles which are the result of entrapped air on the exterior surfaces of the case shall not be considered as leakage.

4.5.6 Vibration. The indicator shall be subjected to a vibration test in accordance with MIL-STD-810, method 514.3, equipment category I, except that curve A shall apply. A standard transmitter circuit shall be adjusted to provide voltages corresponding to an indication of approximately midscale on the indicator. Pointer oscillation shall not exceed 5 percent double amplitude. Upon completion of this test, the indicator shall be subjected to the scale error test at room temperature. The scale errors shall not exceed the tolerances specified in the detail specification.

4.5.7 Low temperature. The indicator shall be subjected to a temperature of -54°C ±2°C for a period of 4 hours. While at this temperature, the indicator shall be tested for scale error and friction as specified in 4.5.3, except in every case, the indicator shall not require more than 5 seconds to come within the specified tolerances. Scale error and friction tolerances shall not exceed the values specified in the detail specification.

4.5.8 High temperature. The indicator shall be subjected to a temperature of 71°C ±5°C for a period of 4 hours. While at this temperature, the indicator shall be tested for scale error and friction as specified in 4.5.3. The scale error and friction tolerances shall not exceed the values specified in the detail specification.

4.5.9 Fogging. The indicator shall be placed in a controlled ambient temperature of 85°C ±5°C for a minimum period of 2 hours. Within 0 seconds after removal from the oven, the indicator shall be immersed to a depth of 1/4 inch in an ice water bath maintained at 0°C to 7°C. After 3 minutes, the indicator shall be removed from the fluid bath and quickly wiped dry (compressed air shall not be used). The cover glass and dial shall immediately be inspected and any evidence of fogging or damage shall be cause for rejection.

4.5.10 Electrical characteristics. The indicator electrical characteristics shall be checked to determine compliance with 3.5.6 through 3.5.6.3.

4.5.11 Magnetic effect. The indicator, first, not operating and second, operating on rated voltage, shall be rotated in a vertical plane about a short bar magnetic compass with the nearest path of the indicator 5-1/2 inches from and magnetically East or West of the center of the compass. Starting directly under the compass, the indicator shall be held in positions 0°, 45°, 90°, 135°, 180°, 225°, 270°, and 315° from the initial position. At each of these positions, the indicator shall be rotated on its own horizontal axis until it is in its normal upright position. The horizontal magnetic field intensity shall be 0.17 to 0.19 oersted. Deflections of the compass at any of the specified positions shall not exceed 1°.

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4.5.12 Effect of lead length. The indicator shall not change more than 1 psi when the lead length between the indicator and transmitter is changed from 5 to 300 feet of No. 20 AWG gage wire. Equivalent resistances may be substituted for the added wire length.

4.5.13 Temperature-altitude. The temperature-altitude test shall be conducted in accordance with MIL-STD-810, method 520, procedure III (except the temperature shall be -55°C) with a chamber pressure equivalent to an altitude of 100,000 feet. The rate of pressure change in the chamber shall not exceed an average of 3 psi per second while under these conditions and at the end of a 1-hour period, the scale error and friction test shall be conducted. The errors shall not exceed the tolerances specified in the detail specification. This test may be conducted concurrently with the low temperature exposure test.

4.5.14 Low temperature exposure. The indicator shall be subjected to a temperature of -65°C ±2°C for a period of 48 hours. While at this temperature, the indicator shall be operated to assure that the pointer moves over the entire scale range without exhibiting sticky or erratic characteristics. The temperature shall then be raised to -55°C ±2°C and held for a period of 24 hours. While at this temperature, the indicator shall be subjected to the scale error and friction test, except that in every case, the indicator shall not require more than 5 seconds to come within the tolerances specified in the detail specification. All parts of the indicator shall be maintained at room temperature for a period of not less than 1 hour and then subjected to the scale error at room temperature test specified in the detail specification. The scale errors shall not exceed the tolerances specified therein and no damage or deterioration shall have occurred that would affect subsequent operation.

4.5.15 High temperature exposure. The indicator shall be subjected to a temperature of 70°C ±2°C for a period of 24 hours. The indicator shall then be tested at this temperature for scale errors as specified in 4.5.3. The scale error and friction tolerances shall not exceed the values specified in the detail specification. All parts of the indicator shall then be maintained at room temperature for a period of not less than 1 hour and the indicator subjected to the scale error at room temperature test specified in the detail specification. The scale errors observed shall not exceed the tolerances specified therein and no damage or deterioration shall have occurred that would affect subsequent operation.

4.5.16 Thermal shock. The indicator shall be subjected to eight cycles of immersion in water at a temperature of 85°C ±2°C and then in water at a temperature of 5°C ±2°C without evidence of moisture penetration or damage to the coating or the case. The length of time for each immersion shall be 30 minutes, with not more than 5 seconds between immersions. After eight continuous cycles have been completed, the indicator shall be subjected to the test specified in 4.5.5. No leakage shall occur as a result of this test.

4.5.17 Salt fog. The indicator shall be subjected to a salt fog test in accordance with MIL-STD-810, method 509.2, for a period of 48 hours and then examined. There shall be no evidence of external corrosion of deterioration that would affect subsequent operation. The indicator shall be subjected to the test specified in 4.5.2. The measured resistance shall be a minimum of 20 megohms. The indicator shall then be subjected to the scale error test. The scale errors shall not exceed the tolerances specified in the detail specification.

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4.5.18 Damping. The indicator pointer shall be positioned at approximately 0-psi pressure. With excitation power off, the test transmitter shall be positioned to the equivalent of 60 psi. Excitation power shall then be applied and the time for the indicator to indicate 60 psi shall not exceed 2.0 seconds. The pointer shall move smoothly, shall not oscillate, and shall not overshoot more than two times.

4.6 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of section 5.

5. PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Unless otherwise specified in the contract or order, indicators shall be individually preserved and packaged in accordance with MIL-P-116, method III, and placed in a snug-fitting unit container conforming to PPP-B-636.

5.1.2 Level C. Indicators shall be individually preserved and packaged in a manner that will afford adequate protection against corrosion, deterioration, or physical damage during shipment from the supply source to the first receiving activity. This level may conform to the supplier's standard practice provided the latter meets the requirements of this level.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. Indicators preserved and packaged as specified in 5.1.1 shall be packed in weather-resistant shipping containers comforming to PPP-B-636. Shipments that exceed the weight limitations of this specification shall be packed in overseas-type containers conforming to PPP-B-601. The gross weight of each PPP-B-601 container shall not exceed 200 pounds. Where practicable, exterior shipping containers shall be of uniform shape and size and of minimum cube and tare. Closure and strapping shall be in accordance with the appendix of PPP-B-636 or PPP-B-601, as applicable. Waterproofing shall be in accordance with MIL-STD-1186.

5.2.2 Level B. Indicators preserved and packaged as specified in 5.1.1 shall be packed in domestic shipping containers conforming to PPP-B-636. Where practicable, exterior shipping containers shall be of uniform shape and size and of minimum cube and tare. The gross weight of each shipping container shall not exceed the weight limitations of the specification. Closure and strapping shall be in accordance with the appendix of PPP-B-636.

5.2.3 Level C. Packages that require overpacking for acceptance by the carrier shall be placed in exterior-type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall comply with Uniform Freight Classification Rules or regulations of other carriers as applicable to the mode of transportation.

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5.3 Physical protection. Cushioning, blocking, bracing, and waterproofing shall be in accordance with MIL-STD-1186. Waterproofing requirements for cushioning materials and containers shall be waived for domestic shipments. The drop tests of MIL-STD-1186 shall be waived when item is preserved, packaged, and packed for immediate use or when the drop tests of MIL-P-116 are applicable.

5.4 Marking. In addition to any other marking required by the order or contract, interior and exterior containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be as specified in the detail specification.

5.4.1 Reinspection date markings. Reinspection date markings shall be in accordance with the instructions of the procuring activity.

6. NOTES

6.1 Intended use. The indicators covered by this specification are intended for use with variable-reluctance type transmitters conforming to MIL-T-25435 in providing remote indication to aircraft pressure functions.

6.2 Ordering data. Ordering data shall be as specified in the detail specification.

6.3 Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Air Force - 11

Review activity:
Air Force - 99, 82

Preparing activity:
Air Force - 11

Project No. 6685-F750

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-25437B	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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